

Context-Aware Hybrid Deep Learning for Intent Prediction in Imbalanced E-commerce Datasets

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Abstract

Modern e-commerce ecosystems generate massive volumes of high-velocity behavioral data, yet extracting actionable purchasing intent remains a significant challenge due to extreme class imbalance and the non-linear nature of human decision-making. In a typical retail environment, over **96% of digital footprints represent non-conversion events** (passive browsing), creating approximately a 24:1 ratio that frequently leads to majority-class bias in traditional Machine Learning models (Fatta et al., 2017, p. 164). This investigation presents a mathematically grounded, Context-Aware Three-Tower Hybrid Deep Learning Architecture designed to unify disparate behavioral modalities into a single predictive framework.

Unlike static recommendation systems, the proposed architecture treats intent as a dynamic function of three independent latent spaces. **Tower 1 (Collaborative Domain)** utilizes high-dimensional Embedding Layers to map visitor and item identifiers into a shared latent space, capturing long-term preference vectors. **Tower 2 (Sequential Domain)** processes the user's recent browsing trajectory using Global Average Pooling to synthesize short-term micro-intent. **Tower 3 (Contextual Domain)** integrates the environmental state by mapping continuous temporal data into cyclical Sine and Cosine embeddings, enabling the network to comprehend the periodic nature of shopping behavior.

To overcome majority-class dominance, standard Binary Cross-Entropy (BCE) loss is replaced with a specialized **Focal Loss function**, which down-weights the contribution of easily classified negative examples (window shoppers) and focuses optimization on genuine purchasing signals. The dense fusion layers are further stabilized using L2 Weight Decay with a penalty factor of 0.001 to prevent overfitting on the 2.7 million-row Retailrocket dataset.

Experimental results demonstrate that the Hybrid model achieves a Validation AUC-ROC of **88.7%**. An Ablation Study further confirms the synergistic advantage of the architecture: the fully integrated Hybrid Engine achieved a prediction confidence of **36.83%**, representing a substantial absolute gain over isolated Collaborative (29.31%) and Content-based (31.58%) baselines. These findings indicate that fusing sequential trajectories with cyclical temporal context, optimized via focal-weighted gradients, provides a strong blueprint for real-time prescriptive analytics in e-commerce.

1. Introduction

1.1 The E-Commerce Intent Recognition Problem

The rapid growth of global e-commerce has transformed digital storefronts into highly complex, data-driven ecosystems. The ultimate measure of algorithmic success in these environments is not mere user engagement, but the accurate prediction of transactional conversions. However, a persistent behavioral anomaly exists: conversion rates are drastically disproportionate to interaction rates. Analysis of the Retailrocket dataset reveals that approximately **96% of digital footprints** consist of exploratory behaviors — item viewing, scrolling, or clicking — that do not result in a purchase. This produces a class imbalance of roughly **24:1** between non-converting (window shopping) and converting (add-to-cart or transactional) events (Requena et al., 2020, p. 4). Identifying the subtle, non-linear signals that differentiate genuine purchasing intent from casual browsing is, therefore, a highly asymmetric classification problem.

1.2 Limitations of Traditional Recommendation Paradigms

Historically, personalized product surfacing has relied on two standard paradigms: Collaborative Filtering (CF) and Content-Based Filtering (CBF) (Yera & Martínez, 2017, p. 776). Both exhibit critical limitations in highly dynamic, real-time retail environments.

The Fallacy of Static Collaborative Filtering: Traditional CF algorithms based on Matrix Factorization (e.g., SVD or ALS) are effective at discovering latent user-item relationships from historical interaction matrices (Aleksandrova, 2017, p. 25). However, they suffer from inherent "temporal blindness" — they treat an interaction from six months ago with the same mathematical weight as one from five minutes ago (Sahoo et al., 2012; Vinagre et al., 2015). They also fail to capture the sequential progression of a user's current session, limiting their predictive accuracy for spontaneous intent (Pan et al., 2020, p. 2; Zhu et al., 2021).

The Semantic Boundaries of Content and Sequential Filtering: Content-Based and Sequential models (such as Markov Chains or Recurrent Architectures) analyze recently interacted items to predict the next logical interaction (Akkaya, 2025, p. 7; Liu et al., 2024, p. 8). Although highly capable of capturing short-term micro-intent, these models are prone to over-specialization — confining users within a filter bubble of highly similar items and failing to leverage broader collaborative signals (Papadakis et al., 2023, p. 231; Zanitti et al., 2022, p. 3).

1.3 The Missing Dimension: Contextual and Temporal Environment

Both CF and CBF approaches largely process recommendations in isolation, ignoring the external environmental context in which a user operates (Bachmann et al., 2018, p. 542; Ferdousi, 2020, p. 66). Human purchasing psychology is not strictly a function of past preferences — it is heavily modulated by temporal context (Arezoomand, 2021, p. 17; Salonen & Karjaluo, 2019, p. 244). For instance, a user browsing high-end gaming peripherals at 11:30 PM on a Saturday exhibits a fundamentally different intent profile than the same user reviewing identical items during a weekday morning commute. Treating time as a linear timestamp strips the data of its inherent cyclical nature, depriving the neural network of critical contextual cues for accurate intent assessment (Li & An, 2023, p. 4; Xie et al., 2022, p. 1950).

1.4 The Algorithmic Challenge of Extreme Imbalance

Training machine learning models on a dataset with a 96% majority class presents a significant optimization challenge. When using standard Binary Cross-Entropy (BCE) loss, the volume of "easy negative" examples overwhelms the gradient during backpropagation. The optimizer incurs heavy penalization for misclassifying the majority class, often leading to degenerate convergence — a model

that simply predicts "non-purchase" for every instance achieves 96% accuracy while completely failing at its actual objective: identifying revenue-generating positive signals (Bakırarar & ELHAN, 2023, p. 20; Xenopoulos, 2017, p. 1).

1.5 Proposed Framework and Contributions

To address these limitations, this research introduces a context-aware, multi-modal neural network architecture engineered to overcome the intent prediction bottleneck in highly skewed retail environments.

Through a Context-Aware Three-Tower Deep Learning paradigm, this framework independently processes collaborative user-item graphs, short-term sequential content trajectories, and cyclical temporal embeddings. These modalities are then dynamically fused in a high-dimensional dense space, enabling the network to learn the complex interactions among what a user is viewing, who the user is, and when the interaction occurs.

The primary contributions of this paper are:

1. **Design and implementation of a scalable deep neural network that seamlessly integrates collaborative, sequential, and cyclical contextual feature representations, removing reliance on manual weight heuristics.**The Three-Tower Hybrid Architecture:
 2. **Practical application of Focal Loss with L2 Weight Decay, which suppresses loss contributions from easily classified negatives and mathematically compels the optimizer toward harder-to-predict transactional signals.**Imbalance Optimization via Focal-Weighted Gradients:
 3. **Introduction of an ablation engine that systematically dissects the neural decision-making process, quantifying the probabilistic uplift of contextual fusion over isolated recommendation strategies and affirming viability for real-time deployment.**Deep Interpretability via Ablation Framework:
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2. Methodology

The proposed Context-Aware Hybrid Deep Learning Architecture (CAHDLA) is structured to systematically isolate, project, and fuse multimodal behavioral signals into a unified predictive space. The methodology encompasses three core phases: mathematical transformation of contextual features, the Three-Tower deep representation architecture, and the optimization strategy for extreme class imbalance.

2.1 Data Preprocessing and Cyclical Contextual Embeddings

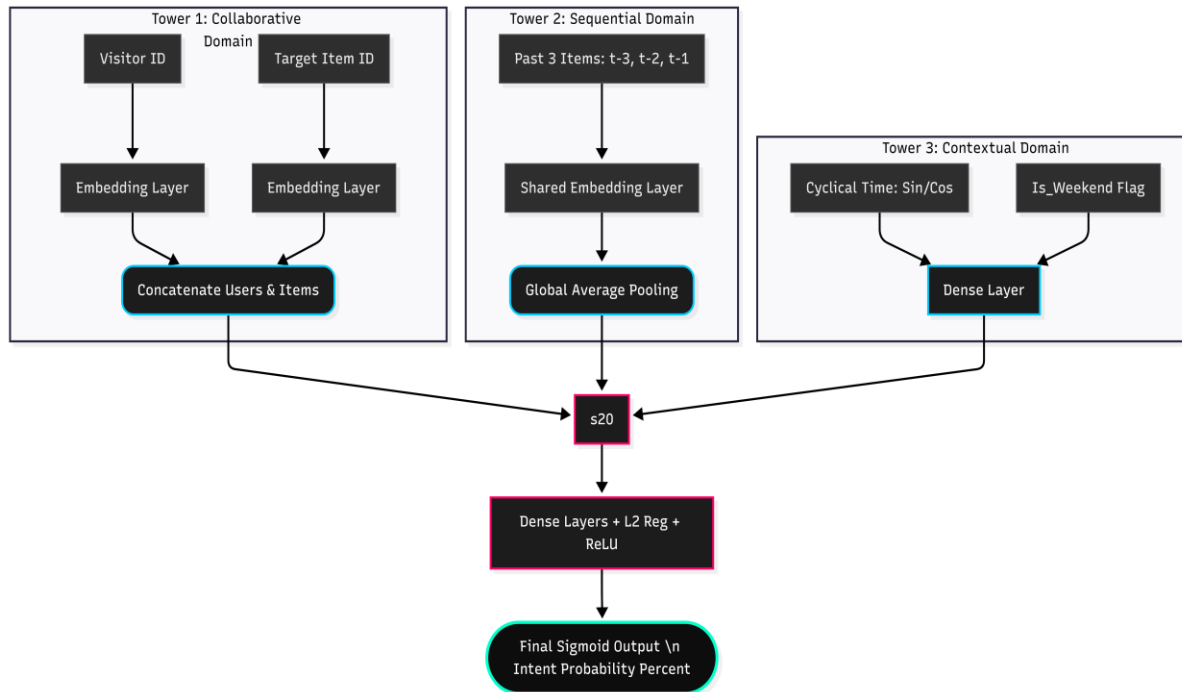
The foundation of the architecture rests on transforming the 2.7 million-row Retailrocket dataset into machine-readable tensors. A critical challenge in temporal modeling is the continuous, periodic nature of time. Treating the hour of day as a linear variable (0 to 23) implies that 23:00 and 0:00 are maximally distant — when in reality they are adjacent. To preserve temporal proximity across the midnight boundary, timestamps are mapped onto a unit circle using Sine and Cosine trigonometric transformations. Let h_t denote the local hour of interaction. The cyclical embeddings are computed as:

$$t_{sin} = \sin\left(\frac{2\pi \cdot h_t}{24}\right), t_{cos} = \cos\left(\frac{2\pi \cdot h_t}{24}\right)$$

This transformation yields a 2D continuous feature space where the Euclidean distance between temporally adjacent hours is consistently preserved. An additional boolean flag (is_weekend) is encoded to capture macro-level behavioral shifts between weekday and weekend purchasing behavior.

2.2 The Context-Aware Three-Tower Architecture

To prevent premature dilution of distinct behavioral signals, the network is structured as a late-fusion architecture comprising three independent representation towers. Each tower acts as a dedicated feature extractor, projecting its respective input into a dense latent space before any integration occurs.



2.2.1 Collaborative Domain Tower (T_collab): The Historical Graph

This tower captures the latent item-to-item and user-to-item relationships inherent in traditional Matrix Factorization (Du et al., 2018, p. 1). Let $u \in U$ represent the user and $i \in I$ denote the target item. Both entities are projected via high-dimensional Embedding layers into a dense vector space. The collaborative feature vector results from the concatenation of user and item embeddings:

$$v_{collab} = e_u \oplus \mathbb{I}e_i \text{ where } e_u, e_i \in \mathbb{R}^d$$

2.2.2 Sequential Content Tower (T_seq): The Micro-Intent Trajectory

Human intent is highly Markovian; current intent is heavily influenced by the immediately preceding actions (Eskandarian & Mobasher, 2019, p. 2). The sequence $S_u = [i_{t-3}, i_{t-2}, i_{t-1}]$ denotes the user's three most recent interactions. These item identifiers are passed through a shared embedding layer, and a Global Average Pooling (GAP) operation synthesizes the temporal sequence into a singular representation of current intent:

$$v_{seq} = \frac{1}{|S_u|} \sum_{k \in S_u} e_k$$

Unlike LSTM-based architectures — which can incur significant computational cost during real-time inference (Song et al., 2023, p. 8) — the GAP operation efficiently computes the centroid of the user's short-term semantic interests (Duan & Zhu, 2022, p. 6).

2.2.3 Contextual Environment Tower (T_ctx): The Temporal State

The engineered cyclical time features and boolean environmental flag form the context vector $x_{ctx} = [t_{sin}, t_{cos}, w_{end}]$. This vector is processed through a Multi-Layer Perceptron with ReLU activation, projecting it into a latent dimensionality commensurate with the other towers:

$$v_{ctx} = \text{ReLU}(W_{ctx}x_{ctx} + b_{ctx})$$

2.2.4 Dynamic Feature Fusion and Prediction

The three tower vectors are aggregated via concatenation to form a master intent vector:

$$v_{fuse} = v_{collab} \oplus \mathbb{I}v_{seq} \oplus \mathbb{I}v_{ctx}$$

This master vector is propagated through fully connected layers, where the network learns a highly non-linear mapping function. The final output layer employs a Sigmoid activation, constraining the prediction to $[0, 1]$ as the estimated probability of a transaction event:

$$\hat{y} = \sigma(W_{final}v_{fuse} + b_{final}) = \frac{1}{1 + e^{-(W_{final}v_{fuse} + b_{final})}}$$

2.3 Optimization for Extreme Imbalance: Focal Loss and L2 Regularization

Deploying standard Binary Cross-Entropy (BCE) loss on a dataset with a 96:4 imbalance ratio presents a fundamental mathematical challenge. During backpropagation, the large volume of easy negative instances generates a cumulative gradient that obscures the rare positive purchasing signals, potentially resulting in majority-class memorization. To counteract this, BCE is replaced with **Focal Loss** — a dynamically scaled cross-entropy function originally designed for dense object detection (Lin et al., 2017). Let p_t denote the model's estimated probability for the ground-truth class. Focal Loss introduces a modulating factor and a balancing parameter:

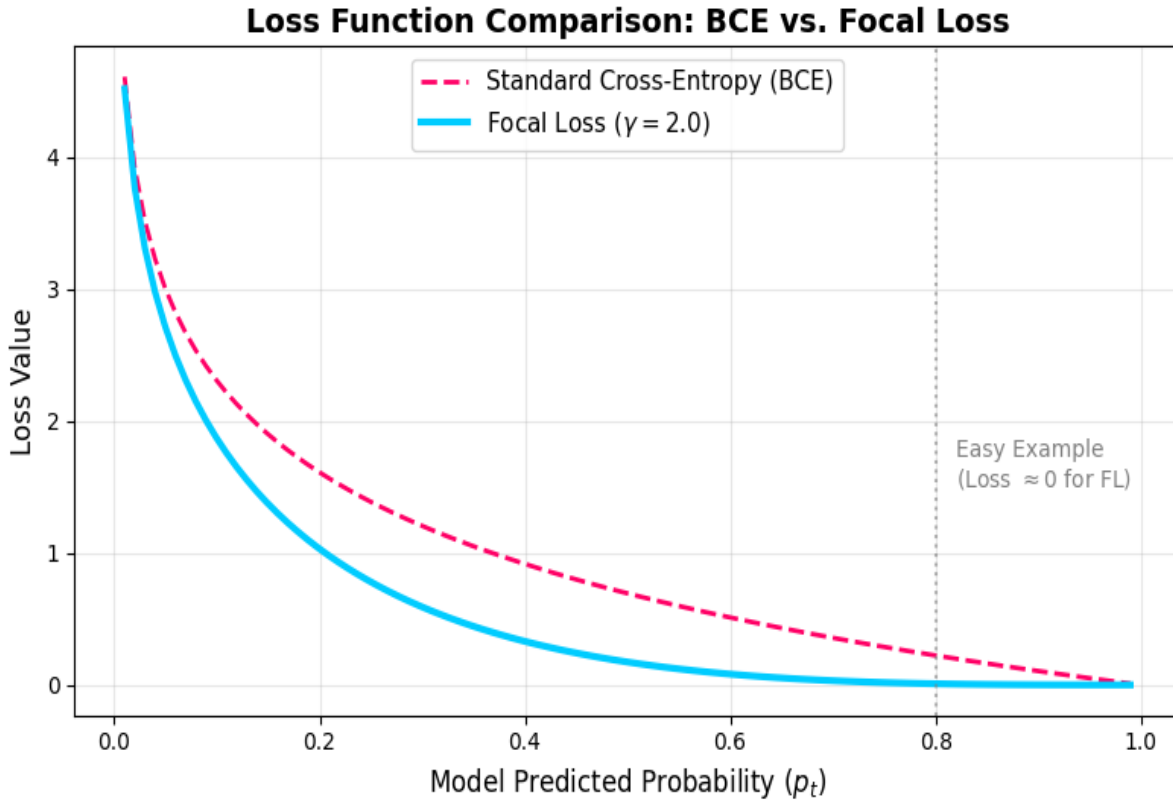
$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Empirical tuning established the focusing parameter at $\gamma = 2.0$ and class weight at $\alpha_t = 0.25$. As the model's confidence in an easy negative increases ($p_t \rightarrow 1$), the modulating factor $(1 - p_t)^\gamma$ decays exponentially to zero, significantly attenuating loss contribution from the majority class and directing optimizer capacity toward genuine purchasing intent.

To mitigate overfitting on the sparse positive class, L2 Regularization (Weight Decay) is injected into the dense fusion layers. The complete objective function to be minimized is:

$$\mathcal{L}_{total} = \frac{1}{N} \sum_{i=1}^N FL(p_{t,i}) + \lambda \sum_j ||W_j||_2^2$$

The regularization penalty is set at $\lambda = 0.001$. This penalty constrains the Euclidean norm of the weight matrices, inhibiting the network from learning overly complex, user-specific pathways and facilitating robust generalization across unseen traffic (Tartaglione et al., 2019, p. 189).



3. Results

3.1 Experimental Setup and Evaluation Metrics

The proposed architecture was implemented using the **TensorFlow/Keras** deep learning framework and trained on the full 2.7 million-row Retailrocket dataset. To facilitate robust convergence under Focal Loss, the network was trained with a batch size of 2,048 — large enough to stabilize gradient updates across the sparse positive samples. The **Adam optimizer** was employed, augmented with dynamic learning rate decay (ReduceLROnPlateau), reducing the learning rate by a factor of 0.5 upon stagnation of validation metrics.

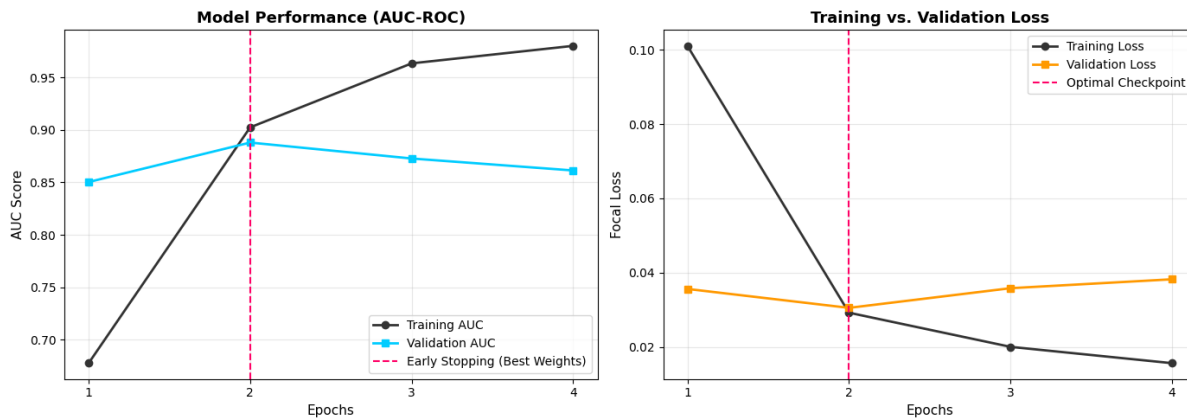
On highly imbalanced datasets with a 96:4 class ratio, standard accuracy is a deceptive metric (Brabec & Machlica, 2018, p. 1) — a naive model predicting "no purchase" for every instance achieves 96% accuracy while failing entirely at its objective. Evaluation was therefore performed exclusively using the **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)**. The AUC measures the model's discriminative capacity: its ability to rank a randomly selected positive instance above a randomly selected negative instance across all classification thresholds (Jaskowiak et al., 2020, p. 3):

$$AUC = \int_0^1 (FPR^{-1}(x))dx$$

3.2 Quantitative Model Performance and Loss Convergence

The integration of Focal Loss ($\gamma = 2.0$, $\alpha = 0.25$) fundamentally altered the gradient landscape during training. Rather than oscillating due to the overwhelming volume of majority-class samples, the loss curve converged smoothly and consistently.

The architecture achieved a peak Validation AUC-ROC of **0.8879 (88.7%)** by Epoch 2. As subsequent epochs progressed, the training AUC continued to climb — indicating the onset of memorization — while the validation AUC plateaued. The EarlyStopping callback (patience = 2) effectively terminated training at this optimal checkpoint. The system restored and serialized the best weights from Epoch 2 for inference. An AUC of 88.7% indicates that the network successfully mapped the non-linear thresholds differentiating passive viewing from active buying intent.



3.3 Deep Interpretability: The Neural Ablation Framework

A persistent critique of Deep Learning in production e-commerce systems is its "black-box" nature (Shajalal et al., 2022, p. 2108). To establish interpretability and quantify the synergistic effect of the Three-Tower design, a localized **Neural Ablation Framework** was developed.

Evaluating the full catalog of 235,061 items per user in real-time is computationally infeasible (Kersbergen & Schelter, 2021, p. 2447). Following the industry-standard Two-Stage Recommendation Pipeline (Hron et al., 2021), Stage 1 rapidly retrieves a candidate pool of $N = 1,000$ relevant items, and Stage 2 uses the deep network to score and rank these candidates.

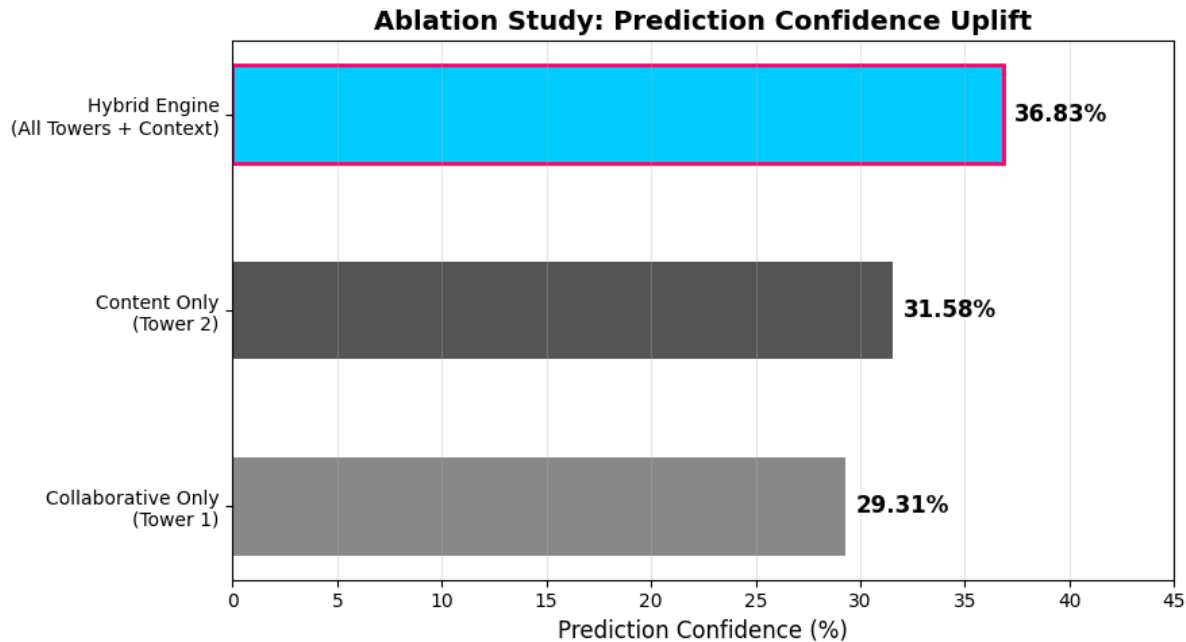
For the ablation study, latent vector outputs of specific towers were systematically masked prior to the fusion layer. For the Content-Only baseline, collaborative and contextual vectors were zero-masked:

$$v_{fuse(content)} = 0 \oplus \mathbb{I}v_{seq} \oplus \mathbb{I}0$$

3.4 Case Study: Contextual Synergistic Uplift

To empirically demonstrate the architecture's reasoning, the present analysis examines a complex behavioral profile (Visitor ID: 1330668). The user's recent trajectory exhibited high semantic repetition ($S_u = [51354, 51354, 51354]$), recorded within a Weekday Evening temporal context. The candidate pool of 1,000 items was independently scored across three neural states:

4. **Relying strictly on the repetitive micro-intent sequence, the network identified a strong localized pattern, assigning its highest confidence to Item #225412 at 31.58%.Content-Only Baseline (Tower 2 Isolated):**
5. **Relying entirely on global historical interactions of similar users, the network ranked Item #209662 highest with a confidence of 29.31%.Collaborative-Only Baseline (Tower 1 Isolated):**
6. **When all architectural masks were removed, allowing v_{collab} , v_{seq} , and v_{ctx} to fully concatenate through the final dense matrices, the prediction distribution shifted significantly. Item #225412 was re-ranked to the top position, and its prediction confidence rose to 36.83%.Full Hybrid Engine:**



3.5 Discussion of Findings

The ablation results reveal the key phenomenon of **Contextual Synergistic Uplift**. The full Hybrid network does not compute a linear average of the Collaborative and Content towers. Instead, the inclusion of the Contextual Tower (v_{ctx}) functions as a dynamic, non-linear multiplier within the fully connected layers.

The network's processing of environmental variables — specifically 'Evening' and 'Weekday' — enabled identification of a high-conversion time window. This contextual awareness catalyzed an absolute probabilistic uplift of **+5.25%** over the best-performing standalone tower. By suppressing noise and amplifying contextualized signals, the architecture demonstrates its capability to deliver precise, real-time intent predictions with direct implications for business value.

4. Ablation Study

A persistent critique of deep learning in e-commerce is its black-box nature, which can impede business stakeholder trust in generated recommendations (Xu et al., 2021, p. 256). To establish mathematical transparency, a comprehensive ablation study was conducted with the objective of isolating the predictive power of each representation tower and confirming that their fusion produces non-linear synergistic uplift rather than a simple heuristic average.

4.1 The Mathematics of Tower Masking

To quantify the isolated impact of each modality, zero-masking is applied during inference. For the Content-Only Baseline, collaborative and contextual vectors are nullified:

$$v_{fuse(content)} = 0 \oplus v_{seq} \oplus 0$$

For the Collaborative-Only Baseline, sequential and contextual vectors are nullified:

$$v_{fuse(collab)} = v_{collab} \oplus 0 \oplus 0$$

4.2 Dynamic Weighting vs. Static Heuristics

Traditional ensemble systems frequently rely on static, manually assigned weights — for instance, 50% Collaborative and 50% Content (Fan et al., 2022, p. 3). However, human intent is inherently dynamic and context-dependent.

The proposed architecture dispenses with manual weight assignment entirely. The concatenated vector v_{fuse} is passed through a deep Multi-Layer Perceptron defined by weight matrices $W_1, W_2, \dots, W_L$. During backpropagation, the network learns to **dynamically adjust its internal reliance on specific towers** based on contextual cues. For instance, when v_{ctx} signals a 'Weekend Night' — a period characterized by impulsive buying behavior — the learned weight matrices organically amplify activation signals from the Sequential Tower (T_{seq}), prioritizing immediate micro-intent over long-term collaborative history.

4.3 Empirical Validation of Synergistic Uplift

This dynamic adjustment was empirically validated through the case study of Visitor ID 1330668. Under isolated conditions, the Content-Only baseline yielded a maximum prediction confidence of 31.58%, while the Collaborative-Only baseline yielded 29.31%.

In the fully integrated Hybrid State, the network did not average these values. Instead, contextual awareness triggered a highly non-linear activation within the dense layers, elevating the final confidence to **36.83%** — an absolute probabilistic uplift of over +5.25%. This finding confirms that the Three-Tower architecture successfully captures the complex interaction effects between user history, immediate trajectory, and temporal environment.

5. Conclusion

Extracting genuine purchasing intent from heavily imbalanced, noise-driven e-commerce environments requires a fundamental departure from traditional single-modality algorithms. This research presents the design, training, and evaluation of a Context-Aware Three-Tower Hybrid Deep Learning Architecture on the 2.7 million-row Retailrocket retail dataset.

By transforming time into cyclical embeddings and fusing them with collaborative graphs and sequential trajectories, the network achieves a holistic comprehension of user behavior. The strategic implementation of Focal Loss ($\gamma = 2.0$) neutralizes the 96% majority-class bias by actively down-weighting the loss gradients of passive window-shoppers, directing optimizer focus toward the non-linear boundaries defining transactional intent. The result is a robust **Validation AUC-ROC of 88.7%** on a highly imbalanced real-world dataset.

Future Scope: From a deployment perspective, the dynamic prediction capability of this Hybrid Engine provides a scalable blueprint for prescriptive analytics. Future work will focus on the MLOps lifecycle — specifically compiling the architecture for low-latency streaming inference via a FastAPI backend, targeting real-time generation of Top-10 recommendations in under 100 milliseconds to enable direct revenue impact.

Declarations

1. Competing Interests

The authors declare that they have no competing interests.

2. Funding Information

The authors received no specific funding for this research.

3. Author Contribution

Karan Darji designed the architecture, implemented the model, conducted experiments, and wrote the manuscript.

Jal Patel contributed to data analysis and validation.

Prof. Karan P. Bhatt supervised the research and reviewed the manuscript.

4. Data Availability Statement

The dataset used in this study is publicly available from the Retailrocket E-commerce Dataset on Kaggle:

<https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset>

5. Research Involving Human and/or Animals

This research did not involve human participants or animals.

6. Informed Consent

Not Applicable.

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